

Sai Samyuktha Paspuleti

Dallas, TX | (469) 783-4804 | saisamyukthapaspuleti@gmail.com | [LinkedIn](#)

Professional Summary

Results-oriented Software Engineer with 3+ years of experience designing scalable backend systems, building modular software components, and integrating cloud-based platforms. Proficient in Python, Java, SQL, and C++, with a strong foundation in data structures, OOP, and system architecture. Skilled in API development, performance optimization, and DevOps practices using CI/CD and Agile. Passionate about delivering robust, efficient, and maintainable code in high-impact applications.

EDUCATION

University of North Texas

August 2023 – May 2025

Master of Science, Computer Science

GPA 3.6

CMR College Of Engineering and Technology

May 2017 – May 2021

Bachelor of Technology, Computer Science Engineering

GPA 3.7

SKILLS

Programming Languages: Python, Java, C++, JavaScript, C, SQL, Shell Scripting, Unix, Linux, PowerShell

Software Development: Object-Oriented Programming (OOP), Data Structures & Algorithms, REST API Design, Multithreading, System Design, Backend Logic Implementation

Web & App Development: Flask (Python), API Integration, Swagger/OpenAPI

Cloud & DevOps Tools: AWS (EC2, S3, Lambda), GitHub Actions, Docker (basic), Jenkins (basic), Git, Amazon Workspace

Databases: MySQL, PostgreSQL, MongoDB, Oracle, SQL Server, Snowflake

Data Tools & Pipelines: Informatica PowerCenter, Informatica Cloud, Apache Airflow, Hadoop (HDFS, Hive, Spark), Databricks

Monitoring & Automation: PowerShell, Bash, Unix CLI, SFTP Integration, Job Scheduling

Analytics & Visualization: Pandas, NumPy, Tableau, Power BI, Matplotlib, Seaborn

Testing & Debugging: PyTest, JUnit, Postman, Log4j, Unit Testing, Error Handling

Methodologies & Collaboration: Agile (Scrum), CI/CD, Sprint Planning, Code Reviews, Jira

Certifications : Certificate of Appreciation in Informatica (Data & Analytics) Infosys, Programming for Everybody – Coursera, Google Professional Data Engineer, Databricks Fundamentals Accreditation – Databricks.

PROFESSIONAL EXPERIENCE

Infosys, Hyderabad, India | System Engineer

June 2021 – August 2023

- Architected and implemented microservice-based modules in Java, improving system modularity and deployment independence.
- Integrated Docker containers into development/test environments, resolving environment drift and reducing setup time by 50%.
- Built and maintained CI/CD pipelines using GitHub Actions and Jenkins, automating code integration, testing, and deployment.
- Enhanced backend service observability using Log4j and ELK stack, reducing mean-time-to-detect by 40%.
- Optimized database indexing and caching, reducing key API response times by 60%.
- Collaborated with cross-functional teams in Agile sprints and mentored 3 junior developers on clean code and API best practices.

Machstatz, Bangalore, India | Software Engineering Intern

June 2020 – May 2021

- Designed and implemented role-based access control in API endpoints, ensuring secure data access and compliance with company policy.
- Built a lightweight REST API in Flask to expose real-time analytics, serving 100+ concurrent clients without performance degradation.
- Wrote integration tests (pytest) and set up automated test runs in CI, increasing code coverage to 85%.
- Collaborated with UX designers to integrate backend services with a React frontend prototype, improving user feedback loops.

PROJECT EXPERIENCE

Scent Recommendation Based on User Mood Inputs, MS (CS)

December 2023-August 2024

- Developed a feature store in Python to preprocess and cache user mood embeddings, reducing inference latency by 30%.
- Implemented A/B testing hooks in the API to measure recommendation quality, resulting in a 12% lift in user engagement.

Credit Scoring Model, MS (CS)

January 2024 –May 2024

- Containerized scoring microservices with Docker and orchestrated with Docker Compose for reproducible local development.
- Exposed a gRPC endpoint for high-speed scoring requests, handling up to 500 TPS in benchmark tests.

Movie Recommendation Service, MS (CS)

August 2024 – December 2024

- Built a caching layer (Redis) to store popular recommendation results, reducing database load by 45%.
- Documented API contracts with OpenAPI/Swagger, enabling seamless integration for frontend and mobile teams.

ORGANIZATIONS

Microsoft Student Ambassador, University of North Texas

December 2023 – May 2024

UNT Artificial Intelligence & Data Science Club – Member

August 2024 – December 2024